

Operating Systems (CS2018)

Final Exam

Date: May 26th 2025

Course Instructor(s)

Dr. Afzal Ahmed

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 5

Roll No

Section

Student Signature

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Attempt all the questions.

CLO #: 1 Identify building blocks of an operating system.

Q1a: State the purpose of block special file and a character special file?

[10 Marks]

Q1b: For each of the following system calls, identify a condition that causes it to fail:

[10 Marks]

- (i) fork,
- (ii) exec,
- (iii) unlink.

CLO #: 2 Compare elements of processes and threads.

Q2: Six batch jobs. A through F, arrive at a computer center at almost the same time. They have estimated running times of 12, 6, 2, 4, 8 and 10 minutes. Their (externally determined) priorities are 3, 6, 2, 1, 4 and 5, respectively, with 6 being the highest priority. For each of the following scheduling algorithms, compute the mean process turnaround time. Ignore process switching overhead.

[20 Marks]

- (i) Round robin.
- (ii) Priority scheduling.
- (iii) First-come, first-served (run in order 12, 6, 2, 4, 8, 10).
- (iv) Shortest job first.

For (i), assume that the system is multi-programmed, and that each job gets its fair share of the CPU. For (ii) through (iv), assume that only one job at a time runs, until it finishes. All jobs are completely CPU bound.

CLO #: 3 Analyze memory management techniques.

Q3a: A computer provides each process with 65,536 bytes of address space divided into pages of 4096 bytes each. A particular program has a text size of 32,768 bytes, a data size of 16,386 bytes, and a stack size of 15,870 bytes. Investigate that will this program fit in the machine's address space? Suppose that instead of 4096 bytes, the page size were 512 bytes, would it then fit? Each page must contain either text, data, or stack, not a mixture of two or three of them.

[10 Marks]

Q3b: A computer has 32-bit virtual addresses and 4-KB pages. The program and data together fit in the lowest page (0–4095). The stack fits in the highest page. Find the entries that are needed in the page table if traditional (one-level) paging is used? Also find that how many page table entries are needed for two-level paging, with 10 bits in each part?

[10 Marks]

CLO #: 4 Appraise characteristics of file systems.

Q4a: The beginning of a free-space bitmap looks like this after the disk partition is first formatted: 1000 0000 0000 0000 (the first block is used by the root directory). The system always searches for free blocks starting at the lowest-numbered block, so after writing file A, which uses six blocks, the bitmap looks like this: 1111 1110 0000 0000. Examine the bitmap after each of the following additional actions:

[10 Marks]

- (a) File B is written, using five blocks.
- (b) File A is deleted.
- (c) File C is written, using eight blocks.
- (d) File B is deleted.

Q4b: Consider a disk that has 10 data blocks starting from block 14 through 23. Let there be two files on the disk: F1 and F2. The directory structure lists that the first data blocks of F1 and F2 are respectively 22 and 16. Given the FAT table entries as below, inspect which data blocks are allotted to F1 and F2?

[10 Marks]

(14,18); (15,17); (16,23); (17,21); (18,20); (19,15); (20, -1); (21, -1); (22,19); 23,14).

In the above notation, (x, y) indicates that the value stored in table entry x points to data block y.

CLO #: 5 Appraise elements of input output techniques.

Q5: Suppose that a system uses DMA for data transfer from disk controller to main memory. Further assume that it takes t_1 nsec on average to acquire the bus and t_2 nsec to transfer one word over the bus ($t_1 \gg t_2$). After the CPU has programmed the DMA controller, find that how long will it take to transfer 1400 words from the disk controller to main memory, if (a) word-at-a-time mode is used, (b) burst mode is used? Assume that commanding the disk controller requires acquiring the bus to send one word and acknowledging a transfer also requires acquiring the bus to send one word.

[20 Marks]